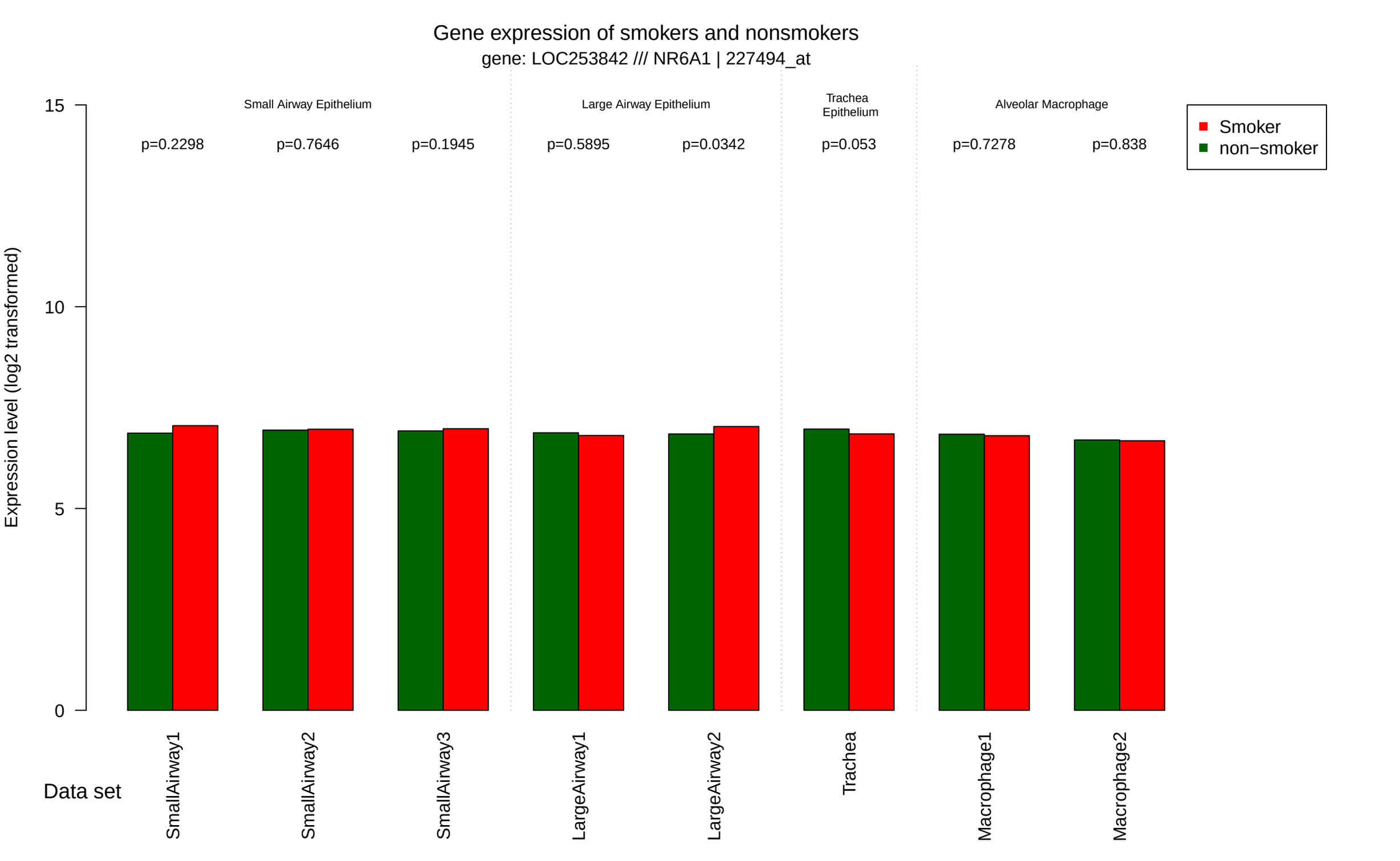
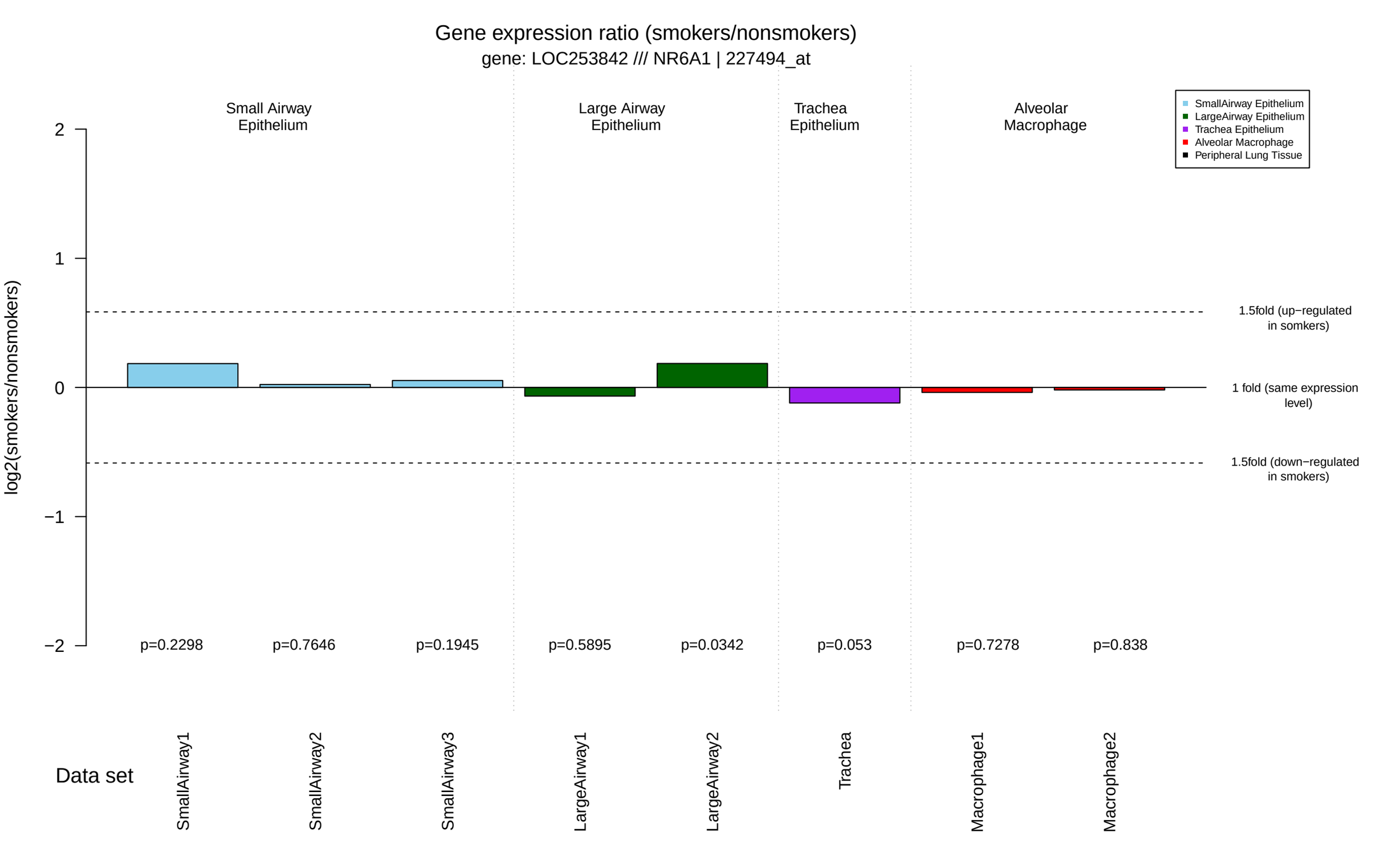
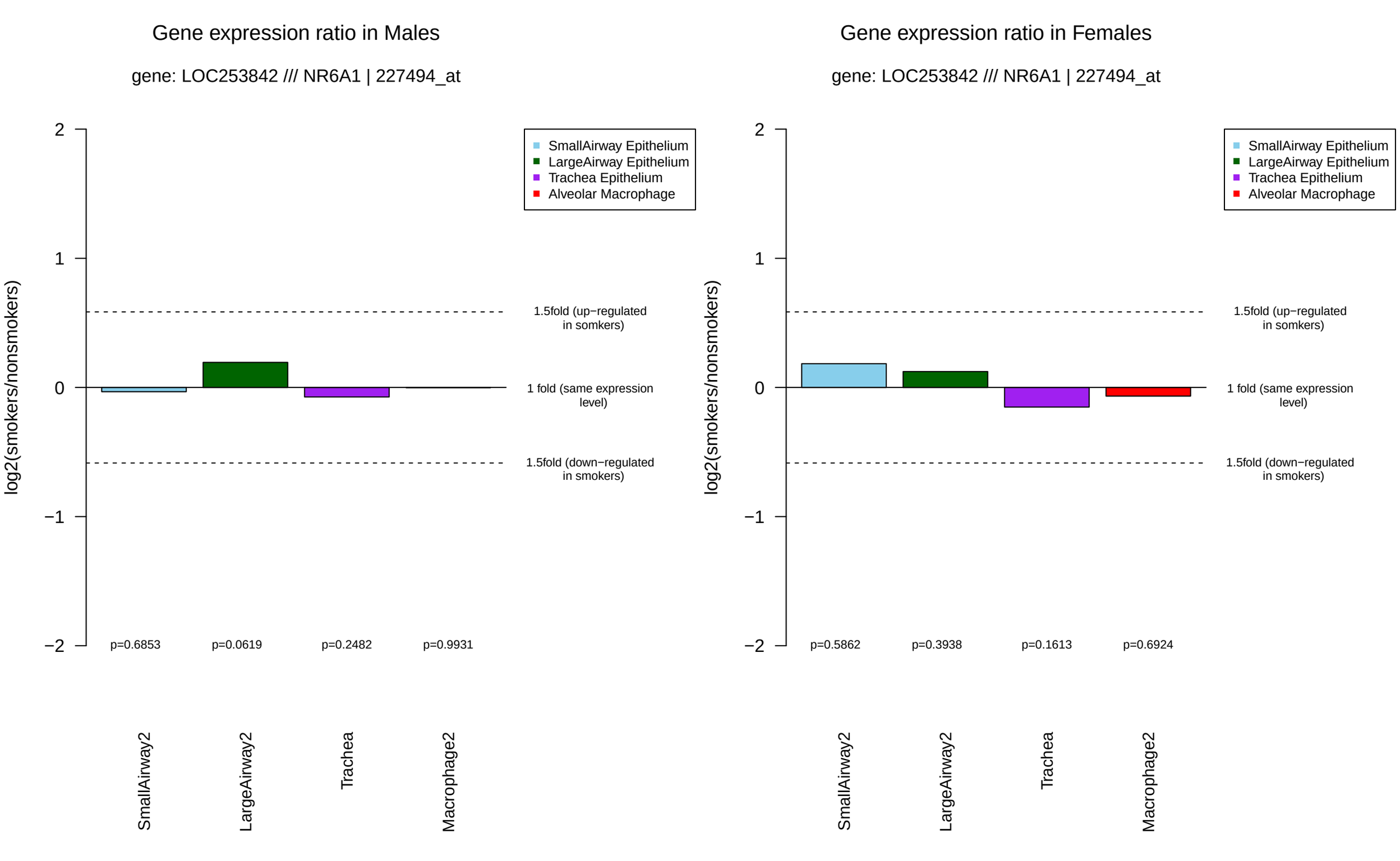
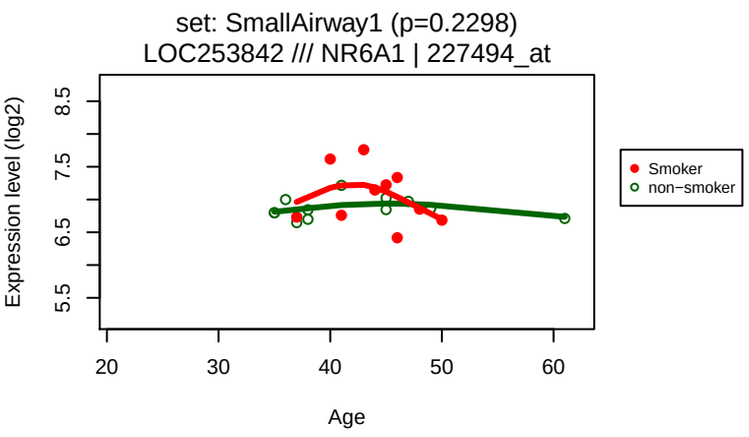
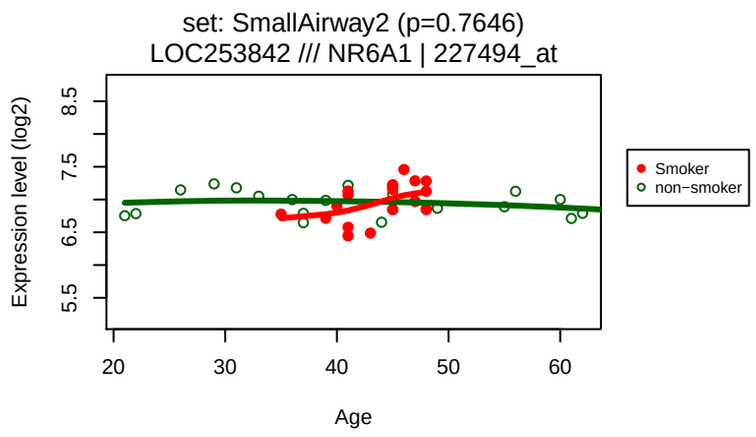
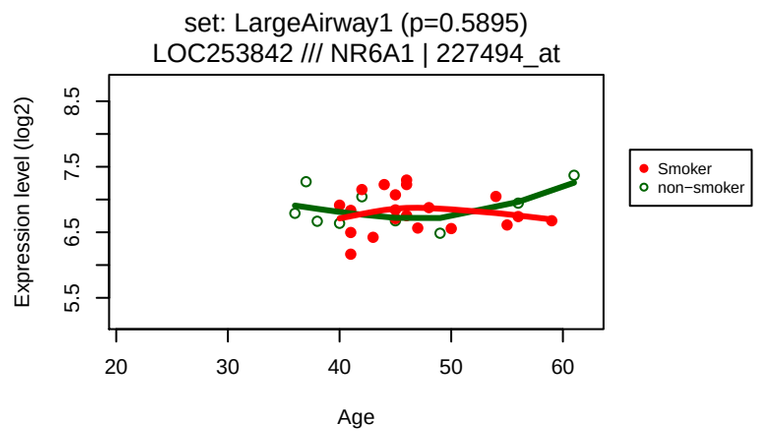

Gene expression of smokers and nonsmokers
gene: LOC253842 /// NR6A1 | 227494_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2298
p=0.7646
p=0.1945
p=0.5895
p=0.0342
p=0.053
p=0.7278
p=0.838
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

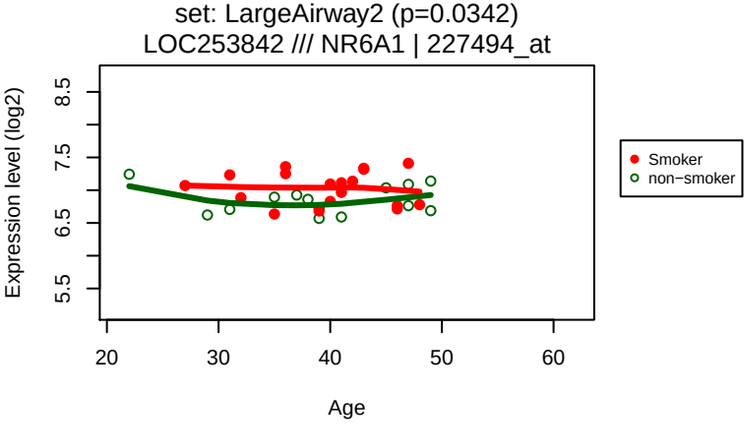

Gene expression ratio (smokers/nonsmokers)
gene: LOC253842 /// NR6A1 | 227494_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2298
p=0.7646
p=0.1945
p=0.5895
p=0.0342
p=0.053
p=0.7278
p=0.838
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

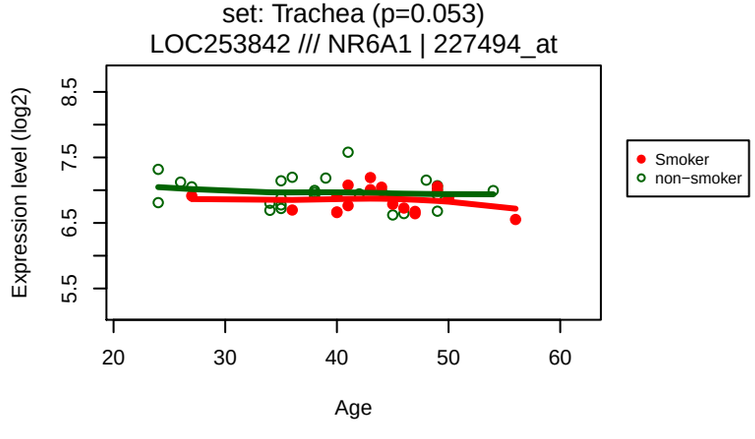

Gene expression ratio in Males
gene: LOC253842 /// NR6A1 | 227494_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6853
p=0.0619
p=0.2482
p=0.9931
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC253842 /// NR6A1 | 227494_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5862
p=0.3938
p=0.1613
p=0.6924
SmallAirway2
LargeAirway2
Trachea
Macrophage2

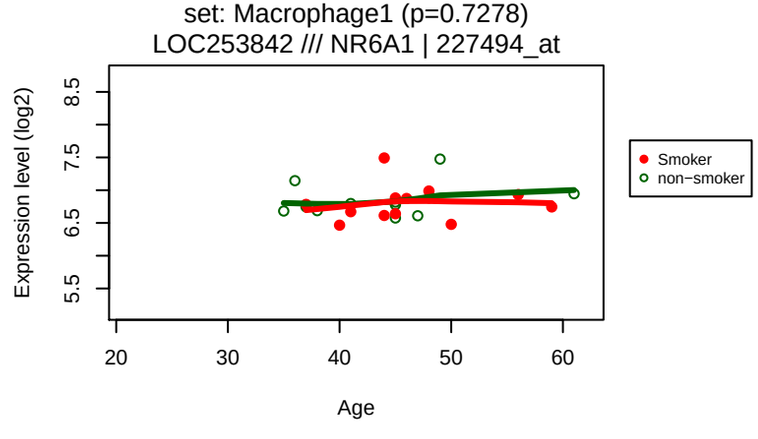

set: SmallAirway1 (p=0.2298)
LOC253842 /// NR6A1 | 227494_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker

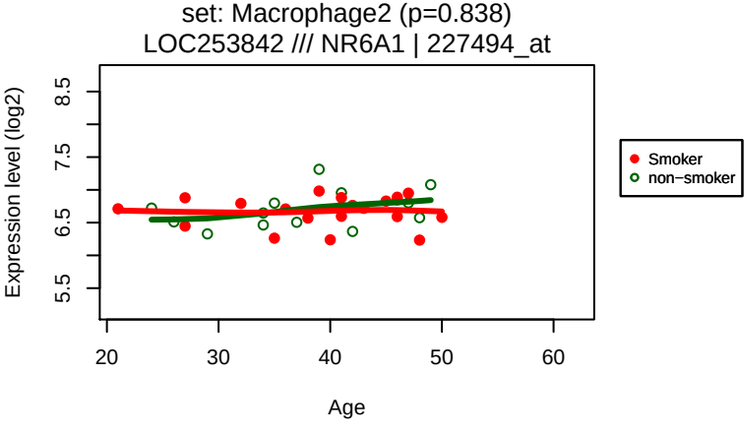

set: SmallAirway2 (p=0.7646)
LOC253842 /// NR6A1 | 227494_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker

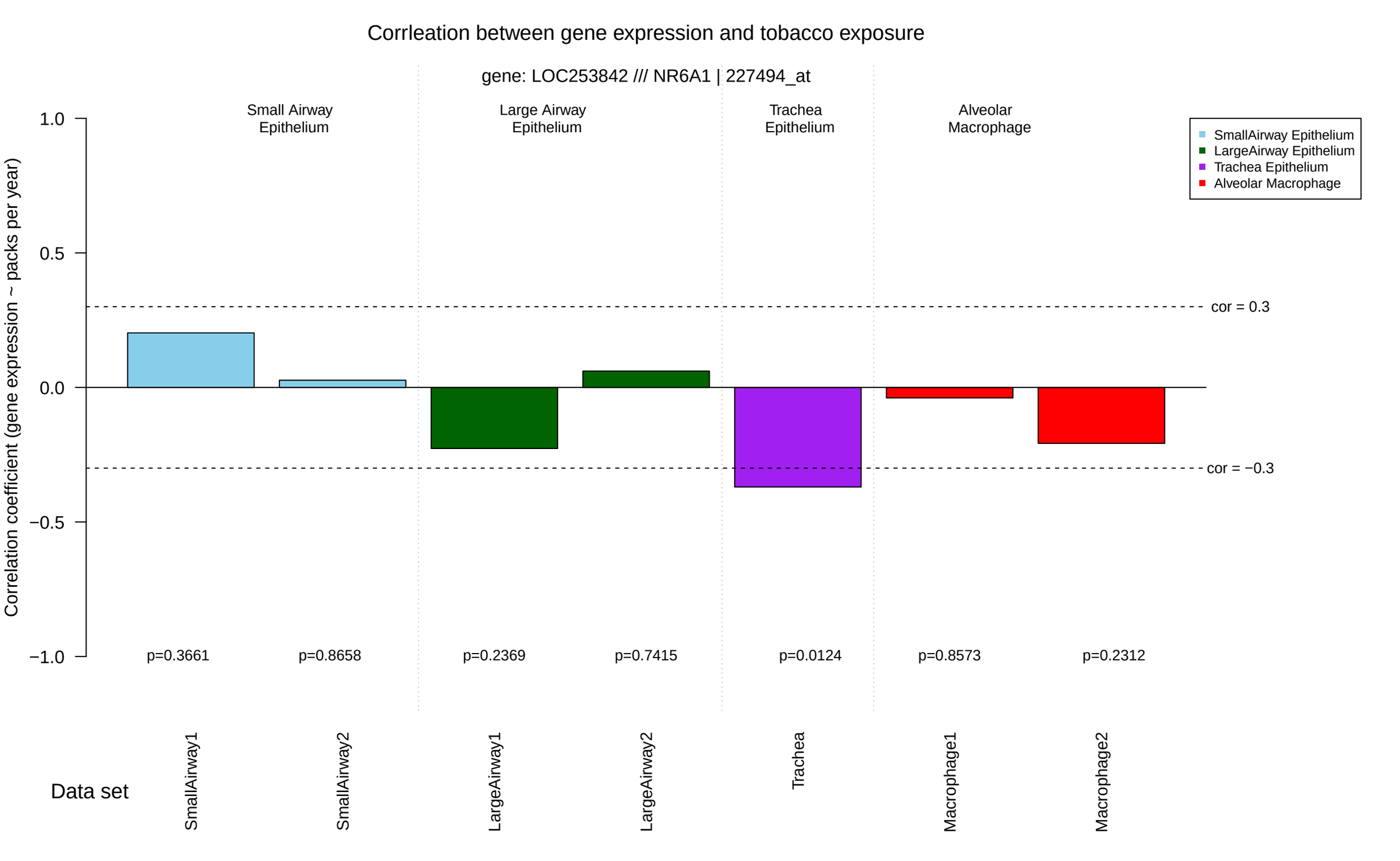

set: LargeAirway1 (p=0.5895)
LOC253842 /// NR6A1 | 227494_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0342)
LOC253842 /// NR6A1 | 227494_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.053)
LOC253842 /// NR6A1 | 227494_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7278)
LOC253842 /// NR6A1 | 227494_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.838)
LOC253842 /// NR6A1 | 227494_at
Expression level (log2)
5.5
6.5
7.5
8.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC253842 /// NR6A1 | 227494_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3661
p=0.8658
p=0.2369
p=0.7415
p=0.0124
p=0.8573
p=0.2312
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2